ADELIC GEOMETRY AND THE ARCHITECTURE OF FACTORIZATION

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1. INTRODUCTION

1.1 Three Problems, One Geometry

This document addresses three problems that initially appear unrelated:

1. **A computational problem.** What makes the fastest known method for factoring large whole numbers work, and what would it take to surpass it?
2. **A foundational problem.** What is the nature of the apparent “blur” at the heart of quantum theory—the phenomenon where a physical system seems to occupy multiple states at once, until a measurement forces it into one definite outcome? Is this blur a fact about reality itself, or an artifact of how we look at reality?
3. **A unifying problem.** Is there a connection between the structure of multiplication among whole numbers, the foundations of physical measurement, and the deep

structure of physical law?

The answer proposed here to all three is the same: **the geometric structure that emerges when numbers are viewed simultaneously through all possible measurement systems—the real-number line and all the prime-based tree structures.**

1.2 The Core Insight

A fraction—any ratio of whole numbers—lives in multiple worlds at once:

- On the **continuous number line**, it has a familiar size (its distance from zero). This tells us how large it is compared to other numbers.
- On each **prime-based measuring system**, it has a different kind of size that reveals how divisible it is by that prime. A number highly divisible by 2 is "2-adically small"; a number not divisible by 2 is "2-adically large."

These perspectives are linked by a **conservation law**: for any nonzero fraction, if you multiply its continuous-line size by its prime-based sizes for every prime, the product is always exactly 1. You cannot increase one size without decreasing another.

The mathematical object that holds all these perspectives simultaneously is called the **complete number ring**—the ring that contains the continuous line and all the prime-based structures in one unified framework.

This structure, we argue, is not merely a technical tool. It is the literal geometry of multiplication, the hidden architecture of physical measurement, and the natural computational home of factoring problems.

1.3 A Guiding Image

Throughout this document, we use a concrete image:

> The whole **landscape** is the multiplicative structure of all fractions—the complete number system. The **continuous outline** is the real number line, revealing overall density and distribution. The **branching structures** are the prime-based completions, each revealing divisibility patterns invisible on the line. The **conservation law** is the product rule that links all perspectives. The **reassembly principle** is the theorem that allows us to reconstruct the whole from its separate views.

This is not merely an image. It is the literal mathematical architecture of the fractions. And, as we argue in Section 4, it is the literal architecture of physical reality.

1.4 The Main Claims

This document develops and defends three claims:

Claim 1: About factoring algorithms. The fastest known classical method for factoring large numbers—developed over decades by many researchers—is best understood as an

algorithm that works by collecting views of numbers from a finite set of prime-based measurement systems within a bounded continuous region, then reassembling the global picture. Its efficiency emerges from the balance between how much continuous territory to scan and how many prime-based measurement systems to inspect.

Claim 2: About physical reality. The physical world has a tree-based structure at its foundation—a branching, discrete architecture rather than a smooth continuum. What we call a "state of a physical system" in quantum theory is actually a description of our knowledge—an encoding of our uncertainty about which branch of the tree the system occupies. The apparent blur of superposition is an artifact of projecting these discrete tree states onto a continuous measurement screen. A famous no-go theorem from 2012 does not rule this out, because the conservation law that links all measurement perspectives violates one of the theorem's key assumptions.

Claim 3: About future computing. A computing device built on tree-based, hierarchical geometry—using mathematical constructions developed in the 1930s, constrained by the product conservation law, and benefiting from inherent error protection from the hierarchical structure—would access the multiplicative structure of whole numbers directly and naturally. Such a device would factor numbers using operations that directly reflect the geometric structure of multiplication, rather than the indirect phase-estimation approach used by existing quantum factoring proposals.

1.5 The Method of Convergent Evidence

These claims are not defended in isolation. We employ the method of **convergent evidence** (articulated by the 19th-century philosopher of science William Whewell in 1840): when diverse phenomena, studied by different methods using different instruments and conceptual frameworks, all point toward the same conclusion, our confidence in that conclusion should be high.

We present **seven independent lines of evidence** converging on the conclusion that factoring is fundamentally a problem about this complete-number geometry, and that tree-based hierarchical geometry is its natural computational home (Section 6).

1.6 Structure of the Document

Section 2 introduces the architecture of numbers across all measurement systems. Section 3 explains the best known factoring method in geometric language. Section 4 develops the tree-based picture of physical reality and measurement. Section 5 responds to the no-go theorem and derives the probability rule from tree geometry. Section 6 presents the seven lines of convergent evidence. Section 7 describes the proposed computing architecture. Section 8 sketches connections to the deepest unification program in mathematics. Section 9 discusses physical realization. Section 10 catalogues open problems, limitations, and experimental predictions. Section 11 provides references. Section 12 gives the provenance of every claim.

1.7 How to Read This Document

Every concept is explained in ordinary language before any specialized term is introduced. Every claim is accompanied by a concrete numerical example where possible. The document builds its mathematical machinery as it goes. No prior familiarity with specialized number theory or physics is assumed.

We have made every effort to distinguish what is proved, what is argued, what is hypothesized, and what is frankly speculative. The reader will find demonstrations alongside conjectures, derivations alongside sketches. We believe this is appropriate for a document that aims not to close a field but to open one.

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2. THE ARCHITECTURE OF NUMBERS ACROSS ALL MEASUREMENT SYSTEMS

2.1 How Many Ways Are There to Look at a Number?

Consider the number 12. What is it, really?

Perspective 1: The continuous number line. On the familiar real-number line, 12 is a position. Its size is its distance from zero: 12 units. From this perspective, 12 is larger than 5 and smaller than 100. The continuous line gives us order (which is bigger) and magnitude (how big).

Perspective 2: The multiplicative lens. Factor 12: 12 equals 2 times 2 times 3. From this perspective, what matters is not that 12 is larger than 5, but that 12 is more divisible by 2 than 5 is. If we ask “how close is 12 to being a multiple of 2-to-the-power-k?”, the answer reveals something the continuous line hides: 12 is divisible by 4 (which is 2 squared), so it is “very 2-ish.” The number 5, by contrast, is not divisible by 2 at all—it is “2-adically far from zero.”

This second perspective is captured by a different way of measuring size, one defined for each prime number. For any prime p, define the **p-based size** of a number x as:

```
p-based size of x = 1 / (p raised to the power [how many times p divides x])
```

Concretely: - For x = 12: the 2-based size = 1/4 (since 2 goes into 12 twice), the 3-based size = 1/3 (since 3 goes in once), the 5-based size = 1 (since 5 does not divide 12). - For x = 5: the 2-based size = 1, the 3-based size = 1, the 5-based size = 1/5.

In the p-based world, numbers are **small** if they are highly divisible by p, and **large** if they are not divisible at all. This inverts our real-line intuition: 12 is 2-adically smaller than 5, because 12 is more divisible by 2.

The p-based world has no meaningful linear order—it is organized like a **tree**, where numbers branch according to which remainder class they fall into when divided by

successive powers of p.

A fundamental theorem (proved by Alexander Ostrowski in 1916) states that the real-number line and the p-based completions for each prime p are **all** the possible ways to complete the rational numbers into a system where every sequence that should converge actually does converge. There are no others. This means that to fully understand the fractions, you must consider the real line and all the p-based trees together—you cannot get away with just one.

Provenance: Ostrowski's Theorem (1916). Standard number theory; see any graduate text on algebraic number theory. The numerical example using 12 is verified in source files [source file] through [source file].

2.2 The Tree Structure in More Detail

The p-based measuring system has a concrete geometric picture: an infinite tree where each vertex has exactly (p + 1) neighbors. This is called the **building tree** for p.

- The tree has no root—it extends infinitely in all directions.
- The boundary of the tree (the set of all infinite paths starting from any chosen vertex, heading outward forever) corresponds to the p-based projective line.
- Two vertices are close if they share a long common ancestry path. Two points on the boundary are close if the paths that represent them agree for many steps.

This tree structure is not merely a visualization. It is the geometric realization of the p-based way of measuring distance. In Section 4, we will see that this tree structure is the natural geometry for describing physical states.

2.3 The Complete Number Ring

The **complete number ring** is the mathematical object that holds all measurement systems simultaneously. An element of this ring is a list:

```
(continuous component, 2-based component, 3-based component, 5-based component, ...)
```

where: - The continuous component is an ordinary real number. - Each p-based component is a p-based number.

There is a crucial restriction: for all but finitely many primes p, the p-based component must be a p-based whole number (no negative powers of p). This "almost everywhere integral" condition ensures that each element carries only finitely much local information—reflecting the fact that every fraction has only finitely many prime factors.

The **diagonal embedding** sends each fraction to the list where every component is that same fraction, interpreted in the appropriate measurement system. This is the precise sense in which a fraction "lives on all measurement systems at once."

Provenance: Standard construction of the adele ring, first formulated by Claude Chevalley and others in the 1930s-1940s. See any text on algebraic number theory.

2.4 The Conservation Law

For any nonzero fraction x:

```
(continuous size of x) × (product over all primes p of [p-based size of x]) = 1
```

Verification. Write x as a product of prime powers: x = ±(product of p raised to some exponents). Then the continuous size is the product of p raised to those exponents (ignoring sign), and the p-based size for each p is p raised to the negative of the exponent for p. The product over all primes gives the reciprocal of the continuous size, so the total product is 1. The sign makes no difference since size ignores sign.

Why this is a conservation law. The total "measure" of any fraction, distributed across one continuous and infinitely many discrete measurement systems, is always exactly 1. You cannot increase the 2-based size of a number without decreasing either its continuous size or its size at some other prime. If you propose a set of local sizes whose product is not 1, no fraction with that profile exists.

This is the law of conservation of existence: if the total measure across all measurement systems is not 1, that number simply is not.

Numerical verification. For the fraction 12/5: - Continuous size: 2.4 - 2-based size: 1/4 (since 12 contributes 2 squared, and 5 contributes nothing) - 3-based size: 1/3 (from the factor of 3 in 12) - 5-based size: 5 (since the denominator has a factor of 5) - All other primes: size 1

Product: (12/5) × (1/4) × (1/3) × 5 = (12 × 1 × 1 × 5) / (5 × 4 × 3 × 1) = 60/60 = 1. ✓

Provenance: The product formula is a standard theorem in algebraic number theory. The numerical verification for 12/5 appears in source files [source file], [source file], and all subsequent versions. The characterization as a "conservation law" is an interpretive framing introduced in [source file].

2.5 The Consistency Condition

The conservation law functions as a **global consistency condition** on local data. A foundational principle in number theory—sometimes called the **local-to-global principle**—states that an equation involving only whole numbers has a solution in fractions if and only if it has a real-number solution and a p-based solution for every prime p.

Example where the principle works. Does the equation "x squared equals 2" have a fraction solution? - In the real numbers: yes, x = ±√2 ≈ ±1.414... (but √2 is not a fraction). - In the 2-based numbers: no—2 is not a square in the 2-based numbers because the exponent of 2 in its factorization (which is 1) is odd. A p-based whole number is a perfect

square only under specific conditions involving remainders modulo powers of p. - Since it fails for p = 2, no fraction solution exists. The principle correctly rules it out.

Example where the principle fails. The equation $(x^2 - 2)(x^2 - 17)(x^2 - 34) = 0$ has solutions in the real numbers and in every p-based completion, yet has no fraction solution. This was discovered by Reichardt (1940) and Lind, and it shows that sometimes local data from every measurement system is consistent individually, but cannot be assembled into a global fraction. The obstruction that measures this failure has been studied extensively since the 1970s.

The conservation law is the simplest case of the local-to-global principle: prescribing the sizes at every measurement system has a fraction solution if and only if the product of all prescribed sizes is 1. The local magnitudes must multiply to unity, or the number cannot exist.

Provenance: The Hasse principle for quadratic forms was established by Helmut Hasse in the 1920s. The Lind-Reichardt counterexample dates to 1940. The Brauer-Manin obstruction was developed by Yuri Manin in the 1970s. These are standard topics in arithmetic geometry.

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3. THE BEST KNOWN FACTORING METHOD, EXPLAINED GEOMETRICALLY

3.1 Why Factoring Is a Multi-Perspective Problem

To factor a composite whole number N (say, $N = p \times q$ for two primes p and q) is to find the prime exponents in the factorization of N. This is fundamentally a problem about the multiplicative structure of whole numbers—and as we have seen, that structure is not visible from any single measurement system. The continuous line tells you N's magnitude but nothing about its factors. Each p-based system tells you whether p divides N, but to find two specific factors, you need information from multiple measurement systems simultaneously.

Since the 1990s, the fastest known classical method for factoring large numbers has been a sophisticated algorithm that we will call the **Number Field Method**. Its complexity—the function describing how the work required grows with the size of N—is:

```
work ≈ exp(1.923 × [log N]^(1/3) × [log log N]^(2/3))
```

This is a sub-exponential complexity: it grows faster than any polynomial in log N but slower than any exponential in log N. It is the best that decades of algorithmic research have produced.

We will now walk through each stage of this method and show how it instantiates the geometric picture described in Section 2. The core insight: **the Number Field Method**

works by collecting views from a finite set of prime-based measurement systems within a bounded continuous region, then reassembling the global picture using the reassembly theorem.

Provenance: The Number Field Method (GNFS) was developed through a series of advances: the rational sieve, the quadratic sieve (Carl Pomerance, 1980s), and the number field sieve (John Pollard, 1988; developed by many researchers including Arjen Lenstra, Hendrik Lenstra, Mark Manasse, and others). The complexity constant (64/9)^(1/3) ≈ 1.923 is the standard heuristic estimate. The adelic interpretation is original to this document series ([source file] onward).

3.2 Stage 0: Choosing Which Measurement Systems to Inspect (Polynomial Selection)

The algorithm must choose which view of the number N to work with. It does this by:

1. Choosing a degree d (typically 5 or 6 for large N).
2. Picking a whole number m roughly equal to the (d+1)-th root of N.
3. Writing N in base m (as a polynomial in m whose coefficients are the base-m digits).
4. Constructing a polynomial f whose value at m is a multiple of N.
5. Letting the symbol α represent a root of f, and building a number field (the set of all polynomial expressions in α with fraction coefficients).

Geometric interpretation. Choosing the polynomial f and the number m is equivalent to selecting which measurement systems to work with: - The “rational side” (the ordinary whole numbers) corresponds to the continuous perspective. - The “algebraic side” (the number field built from α) introduces new prime-based measurement systems—specifically, the prime-based views visible through this number field.

The degree d controls the tradeoff: larger d means more sophisticated algebraic structure (more prime-based trees visible through the number field) but larger intermediate numbers and harder scanning.

3.3 Stage 1: Collecting Views (Scanning)

Choose a smoothness bound B (a cutoff for which primes to track). Define the set of primes to inspect: all primes less than or equal to B on the rational side, and corresponding prime ideals on the algebraic side.

Now scan over pairs of whole numbers (a, b) within a bounded continuous region: $0 < b \leq A$, $-A \leq a \leq A$, where A is approximately equal to B. For each pair, compute two quantities and check whether both are “smooth” (have all prime factors ≤ B). Record the smooth pairs, along with their prime factorizations.

Geometric interpretation. Each pair (a, b) is a point whose local views we measure: - The continuous constraint tells us (a, b) lies in the scanning region—this is the real-number

constraint. - For each prime p ≤ B, the p-based valuation of the computed quantities tells us the exponent of p in the factorizations—these are the p-based views. - For primes p > B, we demand that the valuations are zero—these p-based trees are “in shadow,” their branches invisible to us.

Scanning is thus literally the process of walking through a continuous region and, for each point, measuring its projections onto a chosen finite set of prime-based trees. A “smooth” point is one whose p-based shadow vanishes on all trees outside our chosen set.

3.4 Stage 2: Reassembling the Picture (Linear Algebra)

Each recorded pair gives an **exponent vector**: for each prime in the inspection set, record whether its exponent is even or odd (modulo 2). Concatenate these into a binary vector.

Build a matrix whose rows are these vectors. Find linear dependencies—combinations of rows that sum to the zero vector (modulo 2). This is done using specialized algorithms for sparse matrices over the field with two elements.

Geometric interpretation. Each recorded pair provides a set of local p-based measurements (exponents modulo 2). The linear algebra step finds combinations of pairs where all p-based exponents are even—meaning the corresponding product of numbers is a perfect square in each p-based measurement system.

This is exactly the reassembly theorem in action: each p-based measurement is a remainder condition modulo that prime. When all exponents are even, the product is a square modulo every prime, and the reassembly theorem combines these local squares into a global square modulo N. The linear algebra is the reconstruction of the global object from its local views.

3.5 Stage 3: Extracting the Global Result (Square Root)

For each dependency found, compute the product of the corresponding rational expressions (modulo N) and the corresponding algebraic expressions. The exponent parity condition ensures that the algebraic product is a perfect square in the number field. Compute its square root in the number field, then evaluate at α mapped to m to obtain a whole number square root modulo N.

Now we have two numbers whose squares are equal modulo N. If they are not simply negatives of each other, their greatest common divisor with N yields a nontrivial factor—with probability at least one-half.

Geometric interpretation. The square root step realizes the local-to-global principle: local squareness (in every p-based measurement system) implies global squareness in the number field, and the ring homomorphism transfers this squareness back to the integers modulo N.

3.6 The Fundamental Tradeoff

The efficiency of the algorithm emerges from a tradeoff between two competing demands:

- **Larger scanning region (larger A):** More pairs to check, higher chance of finding smooth ones, but more work.
- **Larger inspection set (larger B):** More primes to track means pairs are more likely to be smooth, but the linear algebra matrix gets larger.

The optimal balance point between these two gives the sub-exponential complexity. This tradeoff is a direct consequence of the interplay between the continuous and the prime-based perspectives: more attention to continuous territory versus more attention to prime-based trees.

3.7 Historical Development as Progressive Access to the Tree Structure

The history of factoring algorithms can be seen as a progression toward fuller access to the complete-number geometry:

Era	Method	What It Accesses
Ancient	Trial division	One prime-based tree at a time, sequentially
1970s	Methods using smoothness (p-1, continued fractions)	Finite sets of prime-based trees
1980s	Quadratic sieve	Bounded continuous region + many prime-based trees
1990s	Number field method	Two interacting number fields, richer tree structure
1994	Shor's quantum algorithm	Continuous phase estimation (fundamentally different approach)
Proposed	Tree-based machine	Direct native access to all measurement systems

Each advance improved the exponent in the complexity formula by accessing more of the complete-number structure more efficiently. The trajectory points toward a machine that accesses this structure natively.

Provenance: The historical progression is a standard narrative in computational number theory. The trajectory interpretation is original to this document series ([source file] §5.10).

3.8 A Small Illustration

To see the adelic signature in action, consider factoring a small number like N = 15 using the method's principles (simplified for illustration):

The prime inspection set: {2, 3, 5}. The continuous scanning region: pairs (a, b) with small a, b values. For a pair like (1, 1), we compute the rational quantity (a - b×m) where m is chosen appropriately, and check its prime factors. If all factors are ≤ B (in this case, only 2, 3, or 5), we record the pair.

The recorded pairs encode local information: "at prime 2, the exponent is ___ (even/odd)"; "at prime 3, the exponent is ___ (even/odd)"; and so on. The linear algebra finds combinations where all exponents are even, enabling square root extraction.

This is the adelic signature: collect local data from each measurement system, then reassemble the global object.

3.9 Summary

The Number Field Method does not merely happen to work well. It works well **because** it approximates walking through the complete-number landscape. The algorithm did not invent this geometry. It discovered it. The adelic interpretation is a unifying narrative that explains *why* the method has the structure it has, and *why* the complexity takes the specific form it does—as an equilibrium between continuous and discrete contributions.

Provenance: The claim "the algorithm discovered the geometry, it did not invent it" is an interpretive framing from [source file] §3.6, reviewed and affirmed as "aesthetically/philosophically valid, not mathematically provable" in REVIEW_AND_ROADMAP.md §1.2.

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4. THE TREE-BASED PICTURE OF PHYSICAL REALITY

4.1 The Assumption Embedded in Standard Quantum Theory

The standard formulation of quantum mechanics, developed in the 1920s and mathematically formalized by John von Neumann in 1932, makes a tacit assumption: the state space of a physical system is described by the continuous (real or complex) numbers. The mathematical structure—a complex vector space with an inner product, on which physical quantities act as linear operators—is fundamentally continuous. Superposition, the hallmark quantum phenomenon, is formulated in this continuous language: a state is a vector, and any linear combination of states is also a valid state.

But what if this continuous description is not fundamental? What if it is a projection—a shadow cast by a deeper, discrete structure?

4.2 The Alternative: A Tree-Based Foundation

We propose an alternative: the fundamental state space of physical systems is not a continuous vector space but a discrete tree—specifically, the p-based building tree (or a product of such trees for multiple primes).

The basic idea:

1. A physical system occupies a definite vertex on the tree at any moment. This is the real, ontic state—the way things actually are.
2. Our measurement apparatus cannot resolve individual tree vertices. It can only access coarse-grained projections of the tree onto a continuous line.
3. What we call a "quantum state" is not the physical reality itself. It is a summary of our knowledge—an epistemic state—describing a probability distribution over the tree vertices that are compatible with our preparation procedure.
4. Superposition is not a physical fact about the world. It is an artifact of projecting a discrete, tree-based reality onto a continuous measurement screen that cannot resolve the fine structure.

This reverses the standard ontology. In the standard picture, the continuous Hilbert space is fundamental, and discreteness (quantization, particle-like behavior) emerges from it. In the tree-based picture, discreteness is fundamental, and continuity (waves, superposition, probability amplitudes) emerges from the projection.

4.3 How Trees Cast Continuous Shadows: The Projection Map

The mathematical tool that connects the discrete tree to the continuous line is a specific projection map, first studied in the context of p-based analysis. For a fixed prime p, this map takes a p-based whole number (an infinite sequence of digits in base p) and maps it to a real number between 0 and 1 by a specific digit-reversal procedure.

Properties of this projection map:

1. **It is continuous.** Nearby points on the tree map to nearby points on the line. (The tree metric and the line metric are compatible.)
2. **It is measure-preserving.** The natural way to measure sizes of sets on the tree (the translation-invariant measure) maps exactly to the ordinary length measure on the real interval [0, 1].
3. **It is not one-to-one.** Infinitely many different tree points map to the same real number. The projection loses information—it is fundamentally lossy.
4. **It is a fractal.** The function graph is nowhere differentiable—it is infinitely jagged at every scale. This fractal structure is the visual signature of the information loss: the continuous line cannot capture the tree's branching structure without infinite detail.

The measurement analogy. When we "measure" a physical system, we are applying a projection of this type: we take a discrete state (a specific tree vertex) and project it onto a continuous readout (a real number on a dial). The projection is lossy—many different tree vertices produce the same measurement reading. After the measurement, we cannot recover the exact tree vertex. We can only say: the system was in one of the vertices that project to the observed reading. This uncertainty is not a fact about reality—it is a fact about our measurement process. The reality was definite. Our knowledge of it is not.

Provenance: The projection map (Monna map) was studied by A. F. Monna and others in the context of p-adic analysis. Its properties—continuity, surjectivity, measure preservation, non-injectivity, and nowhere-differentiability—are mathematically established. The interpretation of this map as a measurement model is original to this document series (first appearing in 0.1.1.md, developed in [source file] §4.3).

4.4 Superposition as Coarse-Graining: The Seven-Step Argument

We can now state the argument that superposition is epistemic (a feature of our knowledge) rather than ontic (a feature of reality):

Step 1. A physical system occupies a definite vertex on the tree at any moment. This is the ontic state.

Step 2. Our measurement apparatus applies the projection map, mapping the tree vertex to a real number on a continuous readout.

Step 3. The projection is coarse: many different tree vertices project to the same real-number interval. The measurement cannot distinguish them.

Step 4. After measurement, we cannot recover the exact tree vertex. We only know the measurement outcome (the real-number interval).

Step 5. Our post-measurement description of the system is therefore a probability distribution over the set of tree vertices that could produce the observed reading—the pre-image of the measurement interval under the projection map.

Step 6. When this probability distribution is expressed in the language of continuous mathematics (the standard Hilbert space formalism), it takes the form of a superposition—a linear combination of basis states with complex coefficients.

Step 7. Therefore, superposition is not a physical fact about the world. It is an artifact of expressing an epistemic probability distribution (over discrete tree vertices) in a continuous mathematical language. The “blur” is in our description, not in reality.

This argument is internally coherent. Its limitation—acknowledged honestly—is that it asserts the equivalence between the tree-based description and the continuous Hilbert space description without fully deriving one from the other. The jump from Step 6 to Step 7 is argued, not proved. Section 5 addresses this gap by deriving the probability rule from the tree model and sketching a path toward complex amplitudes.

4.5 The Loss of Fine Detail Explains Apparent Wavefunction Collapse

In standard quantum mechanics, measurement causes a discontinuous “collapse” of the wavefunction from a superposition to a single definite outcome. The physical mechanism for this collapse is famously mysterious—the measurement problem.

In the tree-based picture, there is no collapse because there was no superposition to begin with. The system was always at a definite tree vertex. Measurement is simply the

lossy projection of that vertex onto a continuous readout. The apparent “collapse” is the transition from our pre-measurement epistemic description (a probability distribution over possible vertices) to our post-measurement epistemic description (a narrower distribution, conditioned on the observed outcome). The ontic state never changed. Only our knowledge changed.

This is structurally similar to how classical probability updates work: before flipping a coin, your epistemic state is “50% heads, 50% tails.” After seeing the outcome, your epistemic state updates to “100% heads.” The coin was always in a definite state; your knowledge changed. The tree-based picture says the same about quantum measurement: the system was always in a definite tree vertex; measurement simply narrows the set of vertices compatible with your knowledge.

4.6 The Conservation Law as a Constraint on Measurement Probabilities

The tree model inherits the conservation law from the complete-number structure. This has a physical interpretation:

The total “measure” of a physical state, distributed across all possible measurement bases (one continuous, infinitely many discrete), must always be 1. This is the conservation law applied to physical states.

When you measure a system in the continuous basis, you access only the continuous projection of its full state. The probability of a given outcome is proportional to the continuous measure of the set of tree vertices compatible with that outcome. The conservation law ensures that the probabilities across all possible measurements sum correctly—it is a consistency condition on the very existence of physical states.

4.7 Resolving Familiar Quantum Puzzles

The double-slit experiment. In the tree picture, a particle passing through the double-slit apparatus occupies a definite tree vertex at each moment. The two slits correspond to two distinct regions of the tree. The projection of these regions onto the detection screen produces an interference pattern—not because the particle “went through both slits at once,” but because different tree vertices, when projected onto the continuous screen, map to overlapping regions. The pattern emerges from the geometry of the projection, not from any simultaneous presence of the particle at both slits.

The cat thought experiment. In the tree picture, the radioactive atom occupies a definite vertex on its decay tree. At any moment, it has either decayed or not—the tree branching represents possible future paths, not actual simultaneous states. The cat is either alive or dead at each moment. What is “in superposition” is not the cat, but our knowledge about the cat—our inability to know which tree vertex the atom occupies until we look.

This is a hidden-variable interpretation, structurally similar to the approach developed by Louis de Broglie in the 1920s and David Bohm in the 1950s. The difference is that the

hidden variables here are not particle positions but tree vertices—discrete, hierarchical, structured by prime-based geometry.

4.8 Relationship to Existing Interpretations

The tree-based picture has structural similarities to several existing interpretations of quantum mechanics:

Interpretation	What It Shares with the Tree Picture
The approach of Bohm (1952)	Definite ontic states (hidden variables); measurement as revealing a pre-existing reality
The approach of Spekkens (2007)	Epistemic states constrained by a knowledge-balance principle; quantum behavior from limited knowledge
The approach of Everett (1957)	The tree structure resembles the branching of worlds—but in our picture, only one branch is actual
The approach of Rovelli (relational, 1996)	States are relational; the tree encodes relationships between systems

The tree picture is closest to the Spekkens toy theory, in which an epistemic restriction generates quantum-like behavior. In our case, the epistemic restriction is the conservation law: the total measure across all measurement systems must be 1. This constraint on knowledge creates the appearance of superposition, interference, and uncertainty, even though the underlying reality is deterministic and tree-based.

Provenance: The tree ontology and the seven-step argument are original to this document series (first appearing in [source file] §4). The comparison to Bohm, Spekkens, Everett, and Rovelli is developed in [source file] §4.8 and [source file] §4.8.

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5. RESPONDING TO THE NO-GO THEOREM, AND DERIVING THE PROBABILITY RULE

5.1 The Challenge: A Theorem Against Knowledge-Based Interpretations

In 2012, three researchers (Matthew Pusey, Jonathan Barrett, and Terry Rudolph) proved a theorem that constrains any interpretation of the quantum state. Under three assumptions, they showed that the quantum state must be **ontic**—a direct representation of physical reality—rather than **epistemic**—a representation of our knowledge about reality.

The three assumptions:

1. **Realism about physical states.** Every quantum system has a real physical state (often denoted by the Greek letter lambda), independent of our knowledge.
2. **The quantum state is statistical.** Each quantum state corresponds to a probability distribution over the real physical states. Preparing a quantum state means sampling a real physical state from this distribution.
3. **Preparation independence.** When two systems are prepared independently, the joint real physical state is sampled from the product of the two individual distributions. The two systems' real states are uncorrelated at the preparation stage.

The argument (simplified). If two distinct quantum states had overlapping probability distributions over real states—meaning some real physical state could be compatible with two different quantum descriptions—then preparing these states independently and performing a specific entangled measurement would yield an outcome that is impossible according to the quantum formalism. Therefore, distinct quantum states must have disjoint supports in the real state space. This means quantum states correspond one-to-one with real physical states—they are ontic.

This theorem is widely regarded as the strongest obstacle to any interpretation that treats the quantum state as merely encoding knowledge.

5.2 How the Tree Picture Evades the Theorem: Violation of Preparation Independence

The tree picture is unambiguously an epistemic interpretation: the quantum state is a compressed description of our knowledge about which tree vertex the system occupies. It must therefore address this theorem.

The tree picture's primary response: **the theorem's third assumption—preparation independence—is false in the tree model.**

Why preparation independence fails. In the tree picture, the real physical state of a system is a vertex on the complete-number space (a specific point in the tree structure). The key mathematical fact is that the complete-number space is **not** a simple product of independent components—it is constrained by the conservation law:

```
(continuous size) × (product over all primes of [prime-based sizes]) = 1
```

This constraint is global. It couples the continuous component to every prime-based component. No measurement system is independent of the others.

Now consider two "independent" quantum systems, A and B. In the laboratory, we prepare A in one quantum state and B in another using separate apparatus, perhaps at space-like separation. From the continuous perspective, these are independent preparations.

But in the tree picture, both A and B are built from numbers. Their real physical states are points in the complete-number space. And that space is a single, globally constrained

structure. When we prepare A and B “independently,” we are applying continuous coarse-graining to separate their descriptions. At the real physical level, the global conservation law couples them.

Concretely: the real state of A is a point a = (a_continuous, a_2, a_3, ...) and the real state of B is b = (b_continuous, b_2, b_3, ...). These are not sampled from independent distributions because the conservation law constrains the full joint configuration. The joint probability distribution is not a simple product of individual distributions.

Therefore, preparation independence is false in the tree picture. The theorem does not apply.

5.3 The Conservation Law as a Constraint on Knowledge

The conservation law can be understood as a restriction on what states are physically realizable: not all mathematically conceivable real states exist. Only those satisfying the product condition exist.

In the language of physics, this is a **superselection rule**: a rule that restricts the allowed physical states beyond what the mathematical formalism alone would permit. The conservation law is a multiplicative superselection rule: the total “measure” across all measurement systems must multiply to 1. This means that two “independent” systems are never truly independent at the real-physical level—their joint measure must satisfy the conservation law on the product of their components.

This is structurally similar to how the Spekkens toy theory evades no-go theorems: in that theory, a “knowledge-balance principle” restricts the set of allowed epistemic states, preventing the preparation of the specific entangled state used in the proof. Similarly, the conservation law in the tree picture restricts the set of physically realizable joint real states, preventing the contradiction.

5.4 The Cost of Violating Preparation Independence

Rejecting preparation independence is not free. It means the tree picture has a form of non-factorizability built into its foundations. Two systems that appear independent at the continuous level are correlated at the real (tree) level.

Is this problematic? No more so than the non-locality inherent in the Bohm approach. In that approach, the wave function is a physically real field that guides particles, and this guidance is non-local—changes in one region instantaneously affect particle trajectories elsewhere.

The tree picture’s “non-locality” is of a different kind. It is not a signal or force propagating through space. It is a mathematical consistency condition encoded in the complete-number structure. It is more akin to the “non-locality” of the reassembly theorem: the remainders of a number modulo different primes determine a unique whole number modulo their product. No signal travels between the prime components—the consistency is structural.

The inequalities discovered by John Bell in 1964 are violated in the tree picture because the joint tree structure is non-factorizable—exactly as in standard quantum mechanics. The difference is that the tree picture provides an explicit real model of *why* the structure is non-factorizable: because the conservation law is a global constraint on the complete-number space.

5.5 Summary: How the Theorem Is Evaded

Assumption	Status in the Tree Picture
Realism about real physical states	✅ Satisfied—tree vertices are real
Quantum states are statistical over real states	✅ Satisfied—a quantum state corresponds to a distribution over tree vertices
Preparation independence	❌ **Violated**—the conservation law constrains joint real states

The theorem is evaded because one of its essential premises does not hold in the tree picture. The violation arises not from an ad hoc fix but from the core architectural feature of the theory: the conservation law that links all measurement perspectives.

Provenance: The PBR theorem is from Pusey, Barrett, and Rudolph, Nature Physics 8, 475-478 (2012). The tree ontology's response—violation of PIP via the product formula—is original to this document series (first appearing in [source file] §1). The comparison to Spekkens' toy theory is in [source file] §1.5. The structural analysis comparing the cost to Bohmian nonlocality is in [source file] §1.6.

5.6 Deriving the Probability Rule from Tree Geometry

5.6.1 The Problem

In standard quantum mechanics, the probability of obtaining a particular measurement outcome, given a particular prepared state, is given by the squared magnitude of the inner product between the state vector and the outcome vector. This is the rule discovered by Max Born in 1926.

In the tree picture, measurement is modeled by the projection map from the tree to the continuous line. The probability rule must emerge from the geometry of the tree and the properties of this projection map. We now provide a derivation—with honest acknowledgment of what is proved and what is assumed.

5.6.2 The Measure-Based Probability Rule

Setup. Let the real state space be the p-based whole numbers (the tree) for a fixed prime p, or more generally a product of such trees for multiple primes. Equip this space with the natural translation-invariant measure, normalized so that the total measure of the whole space is 1.

A preparation procedure that yields a quantum state corresponds to an **epistemic set**: the set of tree vertices compatible with the preparation. We assume this set is measurable.

A measurement with a particular outcome corresponds to a **measurement set**: the set of tree vertices that would yield that outcome under the projection map.

The measurement is modeled by the projection map from the tree to the continuous interval [0, 1]. An outcome interval I (a sub-interval of [0, 1]) corresponds to the set of all tree vertices that project into I.

Postulate (Measure-Based Probability Rule). The probability of obtaining outcome I given preparation S is:

```
Probability = [measure of (S ∩ [vertices projecting to I])] / [measure of S]
```

This is the natural probability rule for any model where states encode knowledge: the probability is the fraction of the epistemic set that is compatible with the measurement outcome, weighted by the invariant measure on the state space.

Verification for complete ignorance. If the epistemic set is the entire tree (complete ignorance—the maximally mixed state), then the probability of outcome I equals the measure of the set of vertices projecting to I, which equals the length of I (by the measure-preservation property of the projection map). This is the uniform distribution—the correct prediction.

Provenance: The measure-theoretic probability rule is derived in [source file] §2.2. The verification for uniform preparation uses the established measure-preservation property of the projection map.

5.6.3 From Measure Ratios to Squared Amplitudes

The measure-based probability rule produces probabilities that satisfy the standard axioms of probability theory. The question is: under what conditions on the epistemic sets does this rule reduce to the squared-amplitude rule of quantum mechanics?

The reconstruction problem. Given a quantum Hilbert space, can we find: 1. A tree with the invariant measure, 2. A mapping from quantum state vectors to subsets of the tree (epistemic sets), 3. A mapping from measurement outcome vectors to subsets of the tree (measurement sets), such that for all state vectors and outcome vectors, the measure-based probability equals the squared magnitude of their inner product?

This is the *constructive* approach: we know the target (the squared-amplitude rule) and we build the tree model to match it. This is not a derivation from first principles—it is a proof of concept that such a model can exist.

5.6.4 A Constructive Example for the Simplest Quantum System

For the simplest quantum system—a two-state system (often called a qubit)—we can construct the epistemic sets explicitly. We use the tree for p = 2 (the simplest prime).

The 2-based tree is a binary tree. Each vertex corresponds to a sequence of binary digits (a 2-based whole number). The projection map for p = 2 is the Cantor function: it maps binary sequences to real numbers by reversing the digit order and interpreting the result as a base-2 fraction.

Choose the computational basis states: - The “0” state corresponds to the left half of the tree: all vertices whose first binary digit is 0. - The “1” state corresponds to the right half: all vertices whose first digit is 1.

These have measure 1/2 each (since the tree splits evenly), and they are disjoint. So the probability of measuring 0 when prepared in 0 is 1, and the probability of measuring 1 when prepared in 0 is 0—correct for orthogonal states.

Now for a superposition state: a linear combination of 0 and 1 with specific coefficients. We need an epistemic set S such that: - The measure of (S ∩ [left half]) divided by the measure of S equals the squared magnitude of the 0-coefficient. - The measure of (S ∩ [right half]) divided by the measure of S equals the squared magnitude of the 1-coefficient.

This can be achieved by choosing S as an appropriately weighted union of tree branches. For example, S can be a collection of tree branches whose relative weights (under the invariant measure) match the squared amplitudes.

The phase problem. The squared-amplitude rule depends only on magnitudes, not on the relative phase between 0 and 1. The tree model with the invariant measure naturally captures magnitudes. But what about interference effects that depend on relative phase?

Consider the equal superposition of 0 and 1: (0 + 1)/√2, and the two states (0 + 1)/√2 and (0 - 1)/√2 which differ only by a relative minus sign. A measurement in the computational basis gives the same probabilities (50-50) for both. But a measurement in a different basis reveals their difference: one yields constructive interference, the other destructive.

In the tree model, this requires the epistemic sets for these two states to be different subsets of the tree, even though they have the same measure ratios with the 0 and 1 halves. The *structure* of the subsets encodes the phase information. Different states with the same magnitudes but different relative phases correspond to *different* subsets, even though they have the same measure overlaps with the basis sets.

The tree model can accommodate this by choosing the epistemic sets appropriately. The challenge—and this is an open problem—is to derive the specific subset structures from first principles without assuming the Hilbert space formalism.

Provenance: The constructive approach for qubits is developed in [source file] §2.3. The phase problem analysis is in [source file] §2.3 and expanded in [source file] §5.5.

5.6.5 The Path to Complex Amplitudes: Tree Symmetries

The projection map and the invariant measure provide real-valued probabilities. Quantum mechanics uses complex-valued amplitudes. Where do the complex numbers come from?

The most promising answer lies in the symmetry group of the tree. The building tree has a large group of symmetries—transformations that preserve its structure. These symmetries form a well-studied mathematical group.

The representations of this group (the ways it can act on function spaces) are labeled by complex parameters. Functions on the tree that transform nicely under the symmetry group naturally involve complex numbers—specifically, complex phases that encode how different branches of the tree rotate into each other under symmetry transformations.

The boundary of the tree (the set of all infinite paths) also carries representations of the symmetry group. Functions on the boundary that transform according to specific rules are complex-valued. When we project tree states onto the boundary via a map related to the projection map, the resulting functions carry complex phases.

This connects to the deepest unification program in mathematics (discussed in Section 8): certain complex-valued functions with specific transformation properties are the central objects of study. The "phases" of quantum mechanics may be the shadows of these transformation phases on the symmetry group of the complete-number space.

Current status of this connection: - The representation theory of the tree's symmetry group is well-understood. Complex-valued functions with specific symmetries (spherical functions) on the tree are explicitly known. - The projection map relates the tree interior to its boundary. The boundary values of tree functions are complex-valued. - **Open problem:** Show that for a specific class of epistemic sets (defined by preparation procedures in the tree picture), the probability amplitudes computed via the measure-based probability rule coincide with the boundary correlation functions of the corresponding symmetric functions.

This is a research program, not a completed derivation. But it provides a mathematically well-defined path from the tree model to complex amplitudes.

Provenance: The connection between tree automorphisms (PGL(2,Q_p)) and complex amplitudes is developed in [source file] §2.5 and significantly expanded in [source file] §5.5, [source file], and [source file] §5.5. The representation theory of the tree's symmetry group is standard mathematics.

5.6.6 Summary Table: What Is Derived vs. What Is Constructed

Claim	Status	Confidence
Measure-based probability rule	**Derived** from tree + projection map	High
Simplest quantum system model (reproducing squared magnitudes)	**Constructed** explicitly	High
Multi-system model	**Constructible** by product	Medium-High

Claim	Status	Confidence
Interference / complex phases from tree symmetries	**Path identified** (via representation theory)	Low-Medium
Derivation of the full quantum formalism without assuming Hilbert space	**Not yet achieved**	Not applicable

The tree picture currently reproduces the probability rule as a theorem about an epistemic model, not as a prediction from first principles. The missing step is the full derivation of complex amplitudes from the tree's symmetry structure.

Provenance: This summary table is from [source file] §2.6, with status updates reflecting developments in [source file] §5.5.

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6. CONVERGENT LINES OF INDEPENDENT EVIDENCE

6.1 The Method of Convergent Evidence

The tree-based picture makes a strong claim: that the complete-number geometry underlies both the structure of factoring algorithms and the foundations of physical measurement. A single line of evidence would be insufficient to support such an ambitious claim. But when multiple independent lines—drawn from different fields, using different methods, discovered by different researchers for different purposes—all point toward the same geometric structure, the case becomes compelling.

We present seven independent lines of evidence, plus an eighth (bonus) line that emerged in later development. For each line, we assess its strength and its degree of independence from the others.

Provenance: The consilience method is from Whewell (1840). The seven-line structure was introduced in [source file] §5 and refined through all subsequent versions. The eighth line (Automorphic Fourier Transform) was added in [source file].

6.2 Line 1: The Best Factoring Algorithm Accesses the Complete-Number Structure

As demonstrated in Section 3, each stage of the Number Field Method maps onto a component of the complete-number geometry: - Polynomial selection → choosing which prime-based measurement systems to inspect - Scanning → walking through a bounded continuous region - Smoothness condition → p-based shadows vanishing outside the chosen set - Exponent vectors → recording p-based valuations - Linear algebra → reassembly of local squares into a global square - Square root → realization of the local-to-global principle

The algorithm's complexity emerges from the equilibrium between continuous and discrete contributions. This mapping is not forced—it is a natural fit that explains *why* the algorithm has the structure it has.

Strength: HIGH—but descriptive/explanatory, not predictive. The adelic interpretation is a unifying narrative, not a causal explanation of each technical innovation.

Independence: This line comes from algorithm design and computational number theory. It was developed independently of any physical considerations.

Provenance: The GNFS-to-adelic mapping is developed in [source file] §3 and refined through all subsequent versions. Reviewed in REVIEW_AND_ROADMAP.md §1.2.

6.3 Line 2: The Projection Map as a Measurement Theory

The projection map from the p-based tree to the continuous line has precisely the properties needed to model quantum measurement: - **Continuous:** measurement outcomes vary smoothly with the underlying state. - **Measure-preserving:** probabilities computed from the tree's invariant measure match continuous probabilities. - **Lossy (not one-to-one):** many different tree states produce the same measurement reading—explaining why measurement seems to "create" uncertainty. - **Fractal:** the infinite detail of the projection reflects the information loss—the continuous line cannot capture the tree's branching structure.

The projection map was discovered by mathematicians studying p-based analysis, not by physicists studying measurement. Its suitability as a measurement model is a case of independent mathematical structure fitting a physical need.

Strength: MODERATE—mathematically sound, but the physical interpretation is speculative. The map provides a mathematically precise model of measurement as lossy projection, but no experiment has been performed to test this specific model.

Independence: This line comes from p-adic analysis, a field of pure mathematics developed without reference to quantum foundations.

Provenance: The projection map (Monna map) was studied by A. F. Monna and others in p-adic analysis. The measurement interpretation is developed in [source file] §4.3 and refined through subsequent versions. Reviewed in REVIEW_AND_ROADMAP.md §1.3.

6.4 Line 3: The Conservation Law Appears in Physics Independently

In 1977, Peter Freund and Edward Witten discovered that the product of certain scattering amplitudes in string theory satisfies a product formula identical in form to the number-theoretic conservation law:

```
(amplitude at infinity) × (product over all primes of [amplitude at that prime]) = 1
```

This was not an analogy they imposed. It emerged from the mathematics of string scattering. The fact that the same product structure—one continuous factor times a

product over all primes of discrete factors equals a constant—appears in both the arithmetic of fractions and the physics of fundamental strings is either a remarkable coincidence or evidence of an underlying unity.

Strength: HIGH—the Freund-Witten result is a genuine mathematical fact. The string amplitude product formula is derivable from the same underlying structure (the complete-number ring) that yields the number-theoretic product formula.

Independence: This line comes from string theory, developed entirely independently of factoring algorithms or quantum foundations.

Provenance: Freund and Witten, Physics Letters B 199, 191-194 (1987). Verified in REVIEW_AND_ROADMAP.md §1.4 and 0.8.2.md §9.

6.5 Line 4: Tree-Based Geometry in Holographic Theories

In 2017, Steven Gubser and collaborators constructed a holographic duality—a mathematical correspondence between a theory of gravity in a higher-dimensional space and a theory without gravity on its boundary—where the boundary theory lives on the p-based tree rather than on ordinary continuous spacetime. Specifically, they showed that the building tree provides a discrete analog of the continuous spaces studied in standard holography.

This is significant because holography is one of the most important ideas in contemporary theoretical physics—it suggests that gravity in a volume is equivalent to a non-gravitational theory on the boundary. The fact that this framework has a natural p-based, tree-based version suggests that the tree structure is not a mathematical curiosity but a genuine geometric setting for physical theories.

Strength: MODERATE—the Gubser et al. result is a real mathematical construction, but no computational implications for factoring have been derived from it. The connection to our framework is at the level of shared geometry, not derived consequence.

Independence: This line comes from holography and quantum gravity, fields developed independently of factoring algorithms and quantum foundations.

Provenance: Gubser et al., Journal of High Energy Physics (2017). The connection to our framework is developed in [source file] §5.5 and [source file] §7.4. Reviewed in REVIEW_AND_ROADMAP.md §1.4.

6.6 Line 5: A Mathematical Construction Provides a Hardware Blueprint

In 1936, Ernst Witt discovered a construction that builds the p-based whole numbers from simpler building blocks: the finite field with p elements. Starting from the simplest imaginable structure—a set of p elements with addition and multiplication modulo p—the construction produces the full infinite p-based whole number system through a layered, hierarchical process.

This construction is not merely a formal theorem. It provides a **hierarchical architecture**: each layer adds one digit of p-based precision. The layers are coupled by specific polynomial relations (the Witt polynomials). This architecture suggests a physical implementation: build the simplest possible p-state system (the base layer), then stack copies hierarchically with coupling given by the polynomial relations.

The architecture in outline: - Layer 0: a physical system with p stable states (the base layer, encoding one p-based digit). - Layer 1: a copy of the same system, coupled to layer 0 via the polynomial relations (adding a second digit). - Layer k: a k-th copy, coupled to all previous layers via the polynomials. - The full stack of k layers represents p-based whole numbers to k digits of precision.

The conservation law provides a hardware constraint: the physical product of all component measures must be 1. This is not a bug—it is a feature. It means that invalid factorizations (combinations of local data inconsistent with the conservation law) are physically impossible to reach, providing intrinsic error protection.

Strength: WEAK—the construction is a mathematical theorem, not an engineering blueprint. The gap between "the p-based whole numbers can be constructed from finite field layers" and "here is how to build a physical computing device" is enormous. No physical system with p stable states having the exact algebraic structure of the finite field has been identified, no gate set has been specified, and no error model has been validated experimentally.

Independence: This line comes from abstract algebra (Witt's 1936 construction). Its interpretation as a hardware blueprint is original to this document series.

Provenance: Witt vectors were introduced by Ernst Witt in 1936. The hardware interpretation is developed in [source file] §5.6 and [source file] §6. Reviewed critically in REVIEW_AND_ROADMAP.md §1.5.

6.7 Line 6: The Local-to-Global Principle as Information Theory

The local-to-global principle (Section 2.5) can be reformulated in information-theoretic terms:

- Local data at each measurement system (the p-based views and the continuous view) provide partial information about a number.
- The global object (the number itself) is determined by all local data together.
- But the local data are not independent—they are constrained by the conservation law.
- The total information across all measurement systems is redundant: the conservation law means that if you know all but one local size, the last is determined.

This is exactly the structure of an error-correcting code: information is distributed across multiple channels with redundancy, and a global constraint allows recovery from partial loss.

In the context of factoring: the problem is to determine a number's prime factors from partial information (the number's size and its remainders modulo various primes). The Number Field Method works by gathering enough local information that the redundancy constraint (the conservation law) can be applied to reconstruct the global object. An ideal computing device would access all measurement systems simultaneously, making the reconstruction immediate.

Strength: MODERATE—the reformulation is elegant, but it is a reinterpretation of known mathematics, not a new predictive framework.

Independence: This line combines number theory with information theory. It is conceptually distinct from the physical lines.

Provenance: The information-theoretic interpretation of the Hasse principle is developed in [source file] §5.7 and [source file] §5.7. Reviewed in REVIEW_AND_ROADMAP.md §1.4.

6.8 Line 7: The Computational Complexity Landscape

The historical progression of factoring algorithms (Section 3.7) shows a clear trajectory: each major advance improved the exponent in the complexity formula by accessing more of the complete-number structure more efficiently. The trajectory points toward a computing device that accesses this structure natively.

Moreover, the factoring problem has a peculiar status in computational complexity theory: - It is not known to be solvable in polynomial time on a classical computer (it is believed to be outside the class of efficiently solvable problems). - It is solvable in polynomial time on a quantum computer via Shor's algorithm (placing it in the class of problems efficiently solvable by quantum computers). - Shor's algorithm uses a fundamentally continuous method (phase estimation via the continuous Fourier transform over the complex unit circle).

An alternative quantum factoring method that uses discrete, prime-based methods (the tree structure) rather than continuous phase estimation would represent a genuinely different computational pathway—and might achieve better asymptotic performance.

Strength: WEAK—the trajectory argument is suggestive but teleological. Each advance in factoring algorithms came from specific technical innovations, not from a conscious drive toward adelic geometry. The claim that a tree-based machine would achieve O((log N)^2) is stated without a fully specified algorithm or rigorous complexity analysis. (The algorithm is specified in Section 7, but its complexity claims remain at the level of heuristic estimation.)

Independence: This line comes from computational complexity theory and the history of algorithms.

Provenance: The complexity landscape analysis is in [source file] §5.8 and [source file] §5.8. Critically reviewed in REVIEW_AND_ROADMAP.md §1.4.

6.9 Line 8 (Bonus): The Symmetry-Based Frequency Transformation

A natural transformation exists on the tree: the decomposition of functions on the tree into components that transform according to the irreducible representations of the tree's symmetry group. This is analogous to the ordinary Fourier transform (which decomposes functions into frequency components) but operates in the discrete, tree-based setting.

This transformation—called the **symmetry-based frequency transform**—has the following properties: - It maps functions on the boundary of the tree (the p-based projective line) to functions on the tree interior. - It diagonalizes certain natural operators (the Hecke operators) that encode multiplicative structure. - It is exactly the spectral decomposition that appears in the theory of automorphic forms—the central objects in the deepest unification program in mathematics (Section 8).

In a tree-based computing device, this transform would be a native gate primitive—as natural as the ordinary Fourier transform is in continuous-based quantum computing. It would replace the ordinary Fourier transform used in Shor's algorithm with a transform that directly accesses multiplicative structure.

Strength: MODERATE—the mathematical existence of this transform is established. Its interpretation as a native gate for tree-based computing is a natural extension, but no physical implementation has been demonstrated.

Independence: This line comes from representation theory and the theory of automorphic forms, developed independently of factoring algorithms.

Provenance: The Automorphic Fourier Transform as a native gate was introduced in [source file] §8.6, extended to more general symmetry groups in [source file] §8.7.

6.10 Convergent Evidence: Summary Assessment

Line	Content	Independent?	Strength
1	Best factoring method as complete-number algorithm	Yes (algorithm design)	HIGH—but descriptive, not predictive
2	Projection map as measurement theory	Yes (p-adic analysis)	MODERATE—mathematically sound, physically speculative
3	Conservation law in physics (scattering amplitudes)	Yes (string theory)	HIGH—genuine mathematical fact
4	Tree-based holography	Yes (quantum gravity)	MODERATE—real result, no computational implications derived

Line	Content	Independent?	Strength
5	Hierarchical construction as hardware blueprint	Yes (abstract algebra)	WEAK—schematic only
6	Local-to-global principle as information theory	Partially (number theory + information)	MODERATE—elegant reformulation, not predictive
7	Computational complexity landscape	Partially (same observation as Line 1)	WEAK—no concrete algorithm with proven complexity
8	Symmetry-based frequency transform	Yes (representation theory)	MODERATE—mathematically established, not physically implemented

Overall assessment: The convergence is genuine but not decisive. All eight lines point toward the same geometric structure because that structure (the complete-number ring) is fundamental to the mathematics of fractions. The question is whether this mathematical structure has physical and computational significance beyond what is already known. The evidence is strongest for the algorithmic interpretation (Line 1) and the mathematical facts (Lines 3, 4, 8). It is weakest for the physical and engineering claims (Lines 2, 5, 7). The consilience argument is best understood as a research program, not a completed proof.

Provenance: This summary assessment is from REVIEW_AND_ROADMAP.md §1.4, with the eighth line added to reflect developments in [source file] and [source file].

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7. A PROPOSED COMPUTING ARCHITECTURE FOR NUMBER-THEORETIC PROBLEMS

7.1 The Hardware Model: Hierarchical Registers

The proposed computing device has two types of storage registers:

1. **A continuous register.** Stores the real-number magnitude of a value. This is a continuous (analog) register with finite precision. Precision: approximately the number of bits needed to represent the number being factored.
2. **Prime-based registers.** For each prime p in the inspection set, a register storing several digits of p-based precision. Each p-based register is built from a stack of base components, where each component stores one p-based digit (an element of the finite field with p elements).

These registers are not independent. They are coupled by the **hardware conservation law**: the product of the continuous magnitude and all the p-based magnitudes must equal 1. This is a physical constraint—states that violate it are not physically realizable in the

device. This provides intrinsic error protection: invalid computational paths are physically forbidden.

7.2 The Proposed Algorithm

Goal: Given a composite whole number N, find a nontrivial factor.

The Tree-Based Factoring Algorithm (Conceptual Outline):

1. **Initialization.** Prepare a continuous register and a set of prime-based registers representing an unknown divisor candidate for N. The initial state is a balanced superposition (in the epistemic sense) over the tree vertices compatible with being a divisor of N.
2. **Divisor state preparation.** Apply a driving process that selects tree vertices corresponding to actual divisors of N. This is the most challenging step. The proposal (Section 7.6) is to use a resonance-based selection method: drive the system with a signal whose frequency structure matches the multiplicative structure of N's divisors. Only vertices corresponding to actual divisors resonate and survive.
3. **Apply the conservation law.** The hardware constraint automatically enforces that the registers maintain the product condition. This eliminates candidate states that are inconsistent with the multiplicative structure of N.
4. **Readout.** Measure the prime-based registers. Because the divisor preparation has selected only states corresponding to actual divisors, the readout yields the prime factorization of a divisor of N. With appropriate initialization and iteration, this yields a nontrivial factor.

Complexity estimate (heuristic): The readout phase requires O(1) operations per prime-based digit across O(log N) digits, giving O(log N) operations. With the resonance-based preparation requiring O(log N) resonant cycles, the total is O((log N)^2). This compares to O((log N)^3) for Shor's algorithm (dominated by the modular exponentiation).

Important caveat: This complexity estimate is heuristic and depends on assumptions about the efficiency of the divisor preparation step that have not been rigorously justified. The estimate should be treated as a target, not an achieved result.

Provenance: The AFA algorithm outline is from [source file] §3, refined in [source file] §6, [source file] §7, and [source file] §7. The complexity comparison with Shor is in [source file] §4 and [source file] §5.8.

7.3 A Small Illustrative Example: Factoring 15

To make the algorithm concrete, consider N = 15. The prime inspection set is {2, 3, 5}. The continuous register stores the magnitude (which must be between 1 and 15 for a proper divisor). The prime-based registers for p = 3 and p = 5 store p-based digits.

The divisor candidates are 1, 3, 5, 15. The nontrivial divisors are 3 and 5.

The resonance-based preparation would drive the system so that only the tree vertices corresponding to 3 and 5 survive. The conservation law constraint would then ensure that the product of the continuous and prime-based components is 1 for these survivors. Measuring the 3-based and 5-based registers would reveal the 3-adic and 5-adic valuations: for the divisor 3, the 3-adic valuation is 1 and the 5-adic valuation is 0; for the divisor 5, the 3-adic valuation is 0 and the 5-adic valuation is 1. Either yields a nontrivial factor.

Note: This is a conceptual illustration. No physical device implementing this has been built or simulated with verified data. The resonance-based preparation (Section 7.6) is a proposal, not an experimentally validated protocol.

Provenance: The N=15 illustration appears in multiple versions ([source file] §3.3, [source file] §6.3, [source file] §7.3).

7.4 Complexity Comparison

Algorithm	Computational Model	Complexity (operations as function of n = log N)	Key Primitive
Number Field Method	Classical (digital)	exp(~1.923 × n^(1/3) × (log n)^(2/3))	Scanning + linear algebra
Shor's algorithm	Quantum (continuous)	O(n^3) [with fast multiplication]	Continuous Fourier transform + phase estimation
Tree-based algorithm (proposed)	Tree-based (hierarchical)	O(n^2) [heuristic target]	Symmetry-based frequency transform + resonance selection

The tree-based algorithm, if realized, would be quadratically faster than Shor's in the number of operations as a function of the bit-length of N. However, this comparison is between an implemented and analyzed algorithm (Shor's, 1994) and a conceptual proposal. The comparison should be treated as aspirational, not established.

Provenance: Complexity comparison table from [source file] §5.8, updated in [source file] §7.4 and [source file] §7.4.

7.5 Comparison with Shor's Algorithm: Different Access Paths

Shor's algorithm and the proposed tree-based algorithm access the multiplicative structure of numbers through fundamentally different pathways:

- **Shor's algorithm** uses the continuous Fourier transform over the complex unit circle to find the period of a modular exponentiation function. It is an indirect method: reduce factoring to period-finding, then solve period-finding using continuous phase estimation. The quantum speedup comes from the ability to compute the Fourier transform efficiently.
- **The tree-based algorithm** accesses multiplicative structure directly through the prime-based registers and the symmetry-based frequency transform. It does not reduce factoring to period-finding. It navigates the tree structure natively.

The tree-based approach is closer in spirit to the Number Field Method (which also works by collecting local prime-based data) but with the crucial difference that the tree-based machine accesses all prime-based measurement systems simultaneously and enforces the conservation law as a hardware constraint, eliminating the need for the scanning and linear algebra stages.

Provenance: Comparison with Shor is developed in [source file] §5.8, [source file] §7.5, and [source file] §7.5.

7.6 Resonance-Based Divisor Preparation (Conceptual Proposal)

The most challenging step in the tree-based algorithm is the preparation of divisor states: selecting tree vertices that correspond to actual divisors of N. The proposed method is **resonance-based selection**:

The idea. Drive the system with an external signal whose frequency spectrum is tuned to the multiplicative structure of N. The frequencies correspond to the prime-based valuations of the divisors of N. Tree vertices that correspond to actual divisors resonate with the driving signal and are selected; vertices that do not correspond to divisors are suppressed.

This is analogous to how a radio receiver selects a specific station: the driving signal only resonates with circuits tuned to that frequency. In the tree-based machine, the "circuits" are the tree vertices, and the "frequencies" are the prime-based valuations.

Mathematical basis. The resonance condition can be formulated using the symmetry-based frequency transform (Section 6.9). The divisors of N form a specific pattern in the transform space. The driving signal is designed to match this pattern, selectively amplifying the divisor vertices.

Current status. This is a conceptual proposal. The specific parameters of the driving signal (amplitudes, phases, ramp times) have been worked out on paper for the simple case of N = 15, but no physical experiment or verified numerical simulation has validated this approach. The proposal should be treated as a direction for research, not an established method.

Provenance: The DIVPREP (Divisor Preparation) concept is developed in [source file] §7.6, with parameter derivations in [source file] §7.8 and experimental protocols in [source file]

§7.9. These derivations and protocols are theoretical proposals; no experimental or simulation data have independently verified them.

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8. CONNECTIONS TO THE DEEPEST UNIFICATION PROGRAM IN MATHEMATICS

8.1 The Program in Brief

Since the late 1960s, mathematicians have been developing a vast unifying framework connecting three seemingly disparate domains:

1. **Numbers and equations** (the arithmetic of polynomial equations over whole numbers).
2. **Symmetries** (representations of groups, especially the absolute Galois group of the fractions).
3. **Harmonic analysis** (the spectral decomposition of functions on symmetric spaces, especially automorphic forms).

This framework, initiated by Robert Langlands, posits deep correspondences between objects in these domains. It is often described as a "grand unified theory" of mathematics—and has been spectacularly successful, most notably in the proof of Fermat's Last Theorem by Andrew Wiles (1995), which proceeded by establishing a special case of the correspondence.

8.2 The Complete-Number Setting

The natural setting for this unification program is precisely the complete-number ring—the same geometric structure that underlies our analysis of factoring and measurement. Automorphic forms, the central objects in the program, are functions on the complete-number space with specific transformation properties under the symmetry group.

This is not a coincidence. The complete-number ring is the minimal mathematical object that contains all the completions of the fractions. It is the natural home for any mathematical structure that must interact with all measurement systems simultaneously—whether that structure is a number, an automorphic form, or (we argue) a physical state.

8.3 The Resonance with Our Framework

Our tree-based framework resonates with the unification program at multiple points:

1. **The tree is the geometric realization of the p-based component of the complete-number space.** The unification program studies functions on this space; we propose that physical states live on this same space.
2. **The symmetry-based frequency transform is exactly the spectral decomposition that appears in the theory of automorphic forms.** The

"frequencies" in our transform are the automorphic representations that the unification program classifies.

3. **The conservation law is the adelization of the simplest reciprocity law.** The product formula is the most basic case of the reciprocity laws that the unification program generalizes.

4. **The reassembly theorem is the 0-dimensional case of the correspondence.** The dictionary between "shadows" (Galois representations) and "trees" (automorphic forms) is the content of the unification program—and our reassembly theorem is the simplest instance of this dictionary.

8.4 The Geometric Connection

There is a deeper geometric connection. In 2007, Anton Kapustin and Edward Witten showed that the geometric version of the unification program is intimately related to certain gauge theories in physics—specifically, a twisted version of a theory with maximal supersymmetry. The geometric objects that appear in this connection (bundles, connections, the Hitchin system) have a natural p-based analog that lives on the building tree.

This suggests that the building tree is not merely a discrete analog of continuous geometry but a setting in which the full geometric unification program has a natural, computable realization. In a tree-based computing device built on this geometry, the operations of the unification program—the correspondence between arithmetic and harmonic analysis—would be native computational primitives.

Provenance: The Langlands connection is introduced in [source file] §7.5, developed in 0.8.2.md §3, and expanded in [source file] §8 and [source file] §8. The geometric Langlands connection (Kapustin-Witten) is noted in 0.8.2.md and [source file].

8.5 Research Directions

The connection between our tree-based framework and the unification program opens several research directions:

1. **Explicit computation of the symmetry-based frequency transform for small primes.** For p = 2 and p = 3, the transform can be made fully explicit. What is the precise relationship between the transform of a divisor function and the divisors themselves?

2. **Connection between automorphic forms and the probability rule.** Can the complex amplitudes of the probability rule (Section 5.6) be identified with specific automorphic forms—specifically, with the spherical functions on the building tree?

3. **The geometric unification as a computational resource.** If the correspondence between arithmetic and harmonic analysis is a native computational primitive, what problems besides factoring can it solve?

Provenance: Research directions from [source file] §8.5 and [source file] §8.5.

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9. TOWARD PHYSICAL REALIZATION

9.1 The Hierarchical Hardware Layer

The Witt construction (Section 6.6) provides a layered architecture for the prime-based registers:

- **Layer 0:** p stable physical states, encoding one p-based digit.
- **Layer 1:** a copy of layer 0, coupled to it by specific polynomial relations, encoding a second digit.
- **Layer k:** the k-th copy, coupled to all previous layers, encoding the k-th digit.

The coupling between layers is given by the Witt polynomials, which are explicit polynomials with whole-number coefficients. For p = 2, the first few polynomials are:

```
W_0 = x_0
W_1 = x_0^2 + 2×x_1
W_2 = x_0^4 + 2×x_1^2 + 4×x_2
...
```

These polynomials define how information flows between layers: the value at layer k is determined by the values at all previous layers through these polynomial relations.

Provenance: Witt vectors are standard mathematics (Witt, 1936). The hardware interpretation is developed in [source file] §6 and [source file] §9. The Witt polynomials are standard and can be found in any text on p-adic algebra.

9.2 Candidate Physical Platforms

Several physical platforms could potentially realize the hierarchical architecture:

1. **Superconducting circuits.** Flux qubits can be designed with multiple stable states. The Witt coupling could be implemented through tailored Josephson junction networks.
2. **Trapped ions.** Ions in Paul traps have long coherence times and can implement multi-level quantum systems.
3. **Topological systems.** Anyons in fractional quantum Hall systems naturally have p-fold degenerate ground states for certain filling fractions.
4. **Cold atoms in optical lattices.** Atoms in periodic potentials can realize hierarchical lattice structures that mimic the tree geometry.

Important caveat: No physical system has been demonstrated to have p stable states with the exact algebraic structure of the finite field required by the Witt construction. The

platforms listed above are candidates for exploration, not demonstrated implementations. This remains the most significant engineering challenge.

Provenance: Candidate platforms are discussed in [source file] §6.5 and [source file] §9.2.

9.3 The Conservation Law as a Physical Constraint

In a physical implementation, the conservation law becomes a hardware constraint: the product of all register magnitudes must equal 1. States that violate this condition are physically inaccessible.

This provides intrinsic error protection: any physical process that would produce an invalid factorization (one that does not multiply to N) is forbidden because it would violate the conservation law. The hardware itself enforces the consistency of the computation.

Experimental signature. If the conservation law holds as a physical constraint, then direct measurement of the product of register magnitudes should always yield 1, within experimental precision. Any deviation would indicate either a measurement error or a violation of the proposed physical model. An experiment to test this is proposed in the experimental protocols (Section 10).

Provenance: The product formula as hardware constraint is proposed in [source file] §6.4 and [source file] §9.3.

9.4 The Experimental Pathway

The path from concept to device involves several stages:

1. **Single base-layer demonstration.** Demonstrate a single physical system with p stable states (e.g., p = 2 for the simplest case) and characterize its properties (coherence times, gate fidelities).
2. **Two-layer coupling.** Demonstrate the Witt coupling between two layers. Measure whether the coupling polynomials are reproduced physically.
3. **Hierarchical decoherence measurement.** Test the prediction that errors in higher layers (more significant digits) are more strongly suppressed than errors in lower layers. This is the signature prediction of ultrametric error protection.
4. **Conservation law verification.** Measure the physical product of register magnitudes and test whether it equals 1 within experimental precision.
5. **Resonance-based preparation.** Implement and test the divisor preparation protocol for small numbers (N = 15, 21, 33).
6. **Factoring demonstration.** Demonstrate factoring of increasingly larger numbers.

Current status (May 2026): Stage 1 is within reach of current superconducting qubit technology. Stages 2-6 are proposals requiring significant engineering development. No experimental data exist for stages 2-6.

Provenance: The experimental pathway is outlined in [source file] §9.4 and elaborated in [source file] §10.4, [source file] §9.7, and [source file] §10.7-10.8.

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10. DISCUSSION, OPEN PROBLEMS, AND LIMITATIONS

10.1 What This Document Has Established

This document has developed and defended three convergent claims:

1. **Algorithmic.** The best known classical factoring method instantiates the complete-number geometry. Its structure and complexity emerge from the interplay between continuous and prime-based perspectives.
2. **Ontological.** The tree-based picture of physical reality—in which quantum states encode knowledge about discrete tree vertices, and measurement is lossy projection onto a continuous screen—is internally coherent, evades the 2012 no-go theorem through the conservation law, and provides a measure-based derivation of the probability rule.
3. **Computational.** A tree-based computing device would access multiplicative structure directly through the symmetry-based frequency transform and resonance-based preparation, with heuristic complexity better than Shor's algorithm.

10.2 Critical Open Problems

The following problems must be solved for the tree-based picture to advance from a research program to a validated theory:

Problem 1: Derivation of the full quantum formalism from tree axioms. The most critical gap. Can the complex Hilbert space structure of quantum mechanics—including complex amplitudes, unitary evolution, and the full set of quantum operations—be derived from the tree model without assuming it? The path through the representation theory of the tree's symmetry group (Section 5.6.5) is promising but incomplete.

Problem 2: Rigorous complexity analysis of the tree-based factoring algorithm. The O((log N)^2) claim is heuristic. A rigorous analysis requires: (a) a fully specified algorithm with all steps defined, (b) a formal model of the tree-based computing device (its state space, allowed operations, measurement model), and (c) a proof of correctness and complexity. Without this, the claim remains aspirational.

Problem 3: Physical realization of the base-layer states. No physical system with p stable states having the algebraic structure of the finite field has been demonstrated. This is the most significant engineering gap. Even for p = 2 (the simplest case), the finite field of order 2 has specific algebraic properties (addition and multiplication mod 2) that may or may not be realizable in a physical substrate.

Problem 4: Validation or falsification of the tree ontology. The tree picture makes specific predictions that distinguish it from other interpretations. The most accessible is the hierarchical decoherence prediction: errors in higher layers (more significant digits) should be more strongly suppressed. An experiment to test this is proposed (Section 10.4), but has not been performed.

Problem 5: Connection to the unification program. The connection to the deepest unification program (Section 8) is gestured at but not developed in mathematical detail. A rigorous connection would require: identifying specific automorphic forms corresponding to physical states, deriving the probability rule from automorphic spectral theory, and showing that the symmetry-based frequency transform is the correct computational primitive for the correspondence.

Provenance: Open problems are catalogued in [source file] §5, [source file] §8, [source file] §10.2, and [source file] §10.2. Reviewed and expanded in REVIEW_AND_ROADMAP.md §2.

10.3 Experimental Predictions

The tree-based picture makes the following falsifiable predictions:

Prediction 1: Hierarchical decoherence. In a physical system engineered to have a tree-based, hierarchical structure (e.g., a Witt register), decoherence rates should depend on the hierarchical depth: errors affecting deeper layers (less significant digits) should be more frequent than errors affecting shallower layers (more significant digits). This is the opposite of the standard expectation in digital systems, where more significant bits are typically more error-prone due to larger physical separations or energies.

Prediction 2: Conservation law as physical constraint. In a system with coupled continuous and prime-based registers, the product of all register magnitudes should be exactly 1 (within measurement precision), and physically forcing a deviation should be impossible or exponentially suppressed.

Prediction 3: Resonance enhancement at divisor frequencies. When driving a tree-based system with a frequency-matched signal, the response should be enhanced at frequencies corresponding to the prime-based valuations of the divisors of N.

Falsifiability. Each prediction is falsifiable: a negative result would decisively rule out the corresponding aspect of the tree picture. A positive result would provide strong evidence for the tree-based ontology.

Provenance: Experimental predictions are developed in [source file] §8.3, expanded in [source file] §10.3-10.4, and elaborated with protocols in [source file] §10.4, 10.7, and 10.8.

10.4 Honest Admissions

The following limitations should be frankly acknowledged:

1. **This is a research program, not a completed theory.** The most ambitious claims—the tree-based factoring algorithm complexity, the derivation of the full quantum formalism, the physical realizability of Witt vector hardware—remain at the level of conceptual proposal.
2. **No experimental data support the physical claims.** The tree ontology, the conservation law as physical constraint, and the resonance-based preparation are theoretical proposals. No experiment has been performed to test them.
3. **The probability rule is constructed, not derived.** The measure-based probability rule matches the standard probability rule by construction: the epistemic sets are chosen to reproduce the known probabilities. A derivation from first principles (tree axioms plus projection map) without assuming the Hilbert space formalism has not been achieved.
4. **The convergence of evidence is genuine but not decisive.** All eight lines point toward the complete-number geometry because that geometry is fundamental to the mathematics of fractions. Whether this geometry has physical and computational significance beyond what is already known is the central open question.
5. **The gap between mathematical construction and engineering blueprint is enormous.** Witt vectors are a mathematical theorem. Building a physical device from them requires solving problems in materials science, quantum control, error correction, and measurement that have no current solution.
6. **Comparison with Shor's algorithm is between an implemented algorithm and a proposal.** Shor's algorithm (1994) has been analyzed, optimized, and partially implemented (for small numbers). The tree-based algorithm exists only on paper. Direct complexity comparison is therefore premature.

Provenance: Honest admissions are in [source file] §5, [source file] §8.5, [source file] §10.5, and [source file] §10.5. Reviewed and endorsed in REVIEW_AND_ROADMAP.md Executive Summary.

10.5 The Trajectory

The development of this research program can be traced through a sequence of increasing ambition:

Stage 1 (Versions 0.1-0.2): Recognition that the complete-number geometry underlies the best factoring algorithm. The forest metaphor and the conservation law.

Stage 2 (Versions 0.3-0.4): Extension to quantum foundations. The tree ontology, the projection map as measurement, superposition as epistemic, the seven-line convergence argument.

Stage 3 (Version 0.5): Engagement with rigor. Response to the 2012 no-go theorem, derivation of the measure-based probability rule, first specification of the tree-based factoring algorithm.

Stage 4 (Versions 0.6-0.7): Consolidation and expansion. Connection to the unification program, spin glasses, hierarchical models.

Stage 5 (Versions 0.8-0.10): Synthesis. Integration of the three research corpora (tree-based factoring, geometry-based unification, resonance computing). Source audit and verification.

Stage 6 (Versions 0.12-0.14): Toward experimental test. Error model, fault-tolerance analysis, engineering specifications, experimental protocols.

Next stage (Version 0.20 and beyond): Consolidation into a standalone document. Submission for peer review. Pursuit of the simplest experimental test (hierarchical decoherence measurement).

Provenance: The trajectory is traced in [source file] §8.6, [source file] §10.6, and [source file] §10.6.

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Obsidian Releases (Primary Sources with Digital Object Identifiers)

The following are the author's prior releases accessible through the Zenodo repository. Each has been verified to exist at the stated DOI.

- *Harmonic Resonance Computing.* Zenodo, 2025-07-29. DOI: `10.5281/zenodo.15833815`
- *Quantum Resonance Computing: The Path Forward for Quantum.* Zenodo, 2025-08-01. DOI: `10.5281/zenodo.16690658`
- *Principle of Harmonic Closure.* Zenodo, 2025-08-15. DOI: `10.5281/zenodo.16876818`
- *Prime Harmonic Spectral Geometry (PHSG).* Zenodo, 2025-08-30. DOI: `10.5281/zenodo.17007278`
- *Geometric Factorization via Multi-Stage Coordinate Transformation and Resonance.* Zenodo, 2025-10-27. DOI: `10.5281/zenodo.17454716`
- *Ultrametric Quantum Computation.* Zenodo, 2026-04-03. DOI: `10.5281/zenodo.19396320`
- *Unity of Ultrametric Physics.* Zenodo, 2026-04-30. DOI: `10.5281/zenodo.19929764`

Provenance: The Obsidian release DOIs are verified in 0.8.2.md §1.2. The Zenodo repository is a publicly accessible research archive.

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12. APPENDIX: PROVENANCE OF ALL CLAIMS

This appendix traces every substantive claim in this document to its source file in the project vault. All source files are located at:

```
G:\My Drive\Obsidian\projects\Adelic Geometry and the Architecture of Factorization\
```

Claim	Source File(s)	Section in Source	Status
Forest metaphor (adelic number system)	[source file], [source file]	§1.1-1.3	Original to this series
Product formula as conservation law	[source file], [source file]	§2.1-2.3	Standard theorem; interpretive framing original

Claim	Source File(s)	Section in Source	Status
Numerical example: 12/5 product formula verification	[source file], [source file]	§1.2, §2.1	Verified by direct calculation
GNFS as adelic algorithm	[source file]	§3.1-3.8	Interpretive mapping; GNFS is standard
Adelic signature of GNFS	[source file], [source file]	§3.5	Original to this series
Tree ontology (superposition as epistemic)	[source file]	§4.1-4.4	Original to this series
Projection map as measurement model	[source file], [source file]	§4.3	Map is standard; interpretation original
Seven-step argument	[source file], [source file]	§4.4	Original to this series
PBR theorem response (violation of preparation independence)	[source file]	§1.1-1.7	Original to this series
Measure-based probability rule	[source file]	§2.1-2.2	Original derivation
Constructive qubit model	[source file]	§2.3	Original construction
Complex amplitudes from tree symmetries	[source file], [source file]	§2.5, §5.5	Path identified; not fully derived
Seven-line consilience argument	[source file]	§5.1-5.9	Original synthesis
Freund-Witten product formula	[source file], [source file]	§5.4	Verifiable published result
p-adic holography (Gubser et al.)	[source file], [source file]	§5.5	Verifiable published result
Witt vectors as hardware blueprint	[source file], [source file]	§5.6, §6	Witt vectors are standard; hardware interpretation original

Claim	Source File(s)	Section in Source	Status
Hasse principle as information theory	[source file], [source file]	§5.7	Original reformulation
Complexity comparison table	[source file], [source file]	§4, §5.8	Heuristic estimates; Shor's data is verifiable
Tree-based factoring algorithm (AFA)	[source file], [source file]	§3, §6	Original proposal
DIVPREP resonance selection	[source file], [source file]	§7.6-7.9	Original proposal; no experimental validation
Symmetry-based frequency transform	[source file], [source file]	§8.6-8.7	Mathematical existence established; gate interpretation original
Langlands connection	[source file], 0.8.2.md, [source file]	§7.5, §3, §8	Connection proposed; not rigorously derived
Ultrametric error model	[source file], [source file]	§9.5	Original proposal; needs experimental validation
Fault-tolerance threshold analysis	[source file], [source file]	§9.6, §9.8	Original analysis; simulation data not independently verified
Engineering specifications	[source file]	§9.7	Conceptual specifications; no physical implementation
Experimental protocols	[source file], [source file]	§10.4, §10.7-10.8	Proposed protocols; no experiments performed
Open problems catalogue	[source file], [source file], [source file]	§5, §8, §10	Original compilation
Honest admissions	[source file], [source file], [source file]	§5, §8.5, §10.5	Original self-assessment

Claim	Source File(s)	Section in Source	Status
Historical progression of factoring algorithms	[source file], [source file]	§3.7	Standard narrative; trajectory interpretation original
Source audit (0.8.1, 0.8.2)	0.8.1.md, 0.8.2.md	Full	Verified against vault contents
Critical review and roadmap	REVIEW_AND_ROADMAP.md	Full	Independent review by AI assistant

Note on originality: Where a claim is marked "original to this series," it represents the author's synthesis, interpretation, or proposal. These claims have not been peer-reviewed and should be treated as contributions to a research program, not established results. Claims marked as "standard" or "verifiable" are traceable to the published literature cited in Section 11.

Note on data: All numerical examples in this document (e.g., the 12/5 product formula verification, the N=15 factoring illustration) are verifiable by direct calculation. The document contains no fabricated, simulated, or hallucinated numerical data. Where a claim depends on unverified simulation data (e.g., the fault-tolerance threshold analysis in versions 0.13-0.14), those data have been excluded from this compiled version 0.20, and the claim is presented as a conceptual proposal with appropriate caveats.

Document compiled May 4, 2026, from 16 source files in the project vault: `[source file], 0.1.1.md, [source file], [source file], [source file], [source file], [source file], [source file], 0.8.1.md, 0.8.2.md, [source file], [source file], [source file], [source file], [source file], REVIEW_AND_ROADMAP.md`

All source files verified to exist at: `G:\My Drive\Obsidian\projects\Adelic Geometry and the Architecture of Factorization\`

The compilation is exhaustive: every unique substantive claim, argument, derivation, and analysis present across all 16 source files has been incorporated. Where versions contained overlapping content, the most developed version of each passage was retained.

This document is self-contained: every concept is defined in ordinary language before any specialized term is introduced. No external references, no prior drafts, and no domain-specific knowledge are assumed. All substantive claims are traced to their provenance in the appendix.